

1. 设 $A \in M_n(\mathbb{C})$, λ_0 是一个特征值, 假设有 3 个关于 λ_0 的 Jordan 链:

$$(A - \lambda_0 I_n)^{m_1-1} v_1, \dots, (A - \lambda_0 I_n) v_1, v_1,$$

$$(A - \lambda_0 I_n)^{m_2-1} v_2, \dots, (A - \lambda_0 I_n) v_2, v_2,$$

$$(A - \lambda_0 I_n)^{m_3-1} v_3, \dots, (A - \lambda_0 I_n) v_3, v_3$$

其中 $m_1 \geq m_2 \geq m_3 \geq 2$, 且

$$(1) (A - \lambda_0 I_n)^{m_1} v_1 = (A - \lambda_0 I_n)^{m_2} v_2 = (A - \lambda_0 I_n)^{m_3} v_3 = 0$$

$$(2) (A - \lambda_0 I_n)^{m_1-1} v_1, (A - \lambda_0 I_n)^{m_2-1} v_2, (A - \lambda_0 I_n)^{m_3-1} v_3$$

线性无关.

证明: 三个 Jordan 链所有向量
线性无关.

试举例说明, 若 v_1, v_2, v_3 线性无关, 三个 Jordan 链所有向量可能线性相关.

(本题说明求过渡阵的 Jordan 链时, 只要保证不同 Jordan 链的链首特征向量线性无关即可)

2. 设 $A, B \in M_n(\mathbb{C})$, $AB=BA$

(1) 设 $s, t \in \mathbb{C}$, 证明: $e^{A(s+t)} = e^{As} e^{At}$

(2) e^A 可逆, $(e^A)^{-1} = e^{-A}$.

(3) 证明: $e^A \cdot e^B = e^B \cdot e^A = e^{A+B}$

3. 设 A 是 n 阶方阵, 求
 $\lim_{k \rightarrow +\infty} A^k$ 存在的充要条件及
极限矩阵.

4. 设 $A \in M_n(\mathbb{C})$, 求证:

$$\sin 2A = 2 \sin A \cos A$$

5. 设 $A \in M_n(\mathbb{C})$, 且 A 可逆.

证明: 存在 $B \in M_n(\mathbb{C})$, $B^2 = A$.

(B 称为 A 的平方根).

举例说明, 若 A 不可逆, 结论不正确.

6. 设 $A \in M_n(\mathbb{C})$, $\text{rank } A = 1$

设 $f(z)$ 在 A 的谱上可定义

证明: 存在一个一次多项式 $P(z)$.

$$P(z), \quad P(A) = f(A).$$

7. 是否存在 $A \in M_2(\mathbb{R})$,

$$\text{使得 } \sin A = \begin{pmatrix} 1 & 2020 \\ 0 & 1 \end{pmatrix}?$$

1. 证明: 令 $i=1,2,3$. $(A-\lambda_0 I_n)^{m_i-k} v_i = \beta_{i,k}$
三个 Jordan 链: $(A-\lambda_0 I_n)^k \beta_{i,k} = 0$.

$$\beta_{11}, \dots, \beta_{1,m_1}, \beta_{21}, \dots, \beta_{2,m_2}, \\ \beta_{31}, \dots, \beta_{3,m_3}. \quad (m_1 \geq m_2 \geq m_3)$$

因为 $\beta_{11}, \beta_{21}, \beta_{31}$ 线性无关,

则 $\beta_{11}, \beta_{12}, \beta_{21}, \beta_{22}, \beta_{31}, \beta_{32}$

线性无关, 因为

$$C_{11}\beta_{11} + C_{12}\beta_{12} + C_{21}\beta_{21} + C_{22}\beta_{22} \\ + C_{31}\beta_{31} + C_{32}\beta_{32} = 0$$

两边左乘 $(A - \lambda_0 I_n)$, 得

$$C_{12} \beta_{11} + C_{22} \beta_{21} + C_{32} \beta_{31} = 0$$

$$\Rightarrow C_{12} = C_{22} = C_{32} = 0$$

$$\Rightarrow C_{11} = C_{21} = C_{31} = 0$$

引入记号:

给定一个链 C_v

$$(A - \lambda_0 I_n)^k v, \dots, (A - \lambda_0 I_n)v, v.$$

$$\text{定义 } \tilde{C}_v = \begin{cases} C_w & \text{若 } v = (A - \lambda_0 I_n)w \\ C_v & \text{若 } v \notin C(A - \lambda_0 I_n) \end{cases}$$

现在假设

$$\beta_{11}, \dots, \beta_{1k}$$

$$\beta_{21}, \dots, \beta_{2k} \quad \text{线性无关}$$

$$\beta_{31}, \dots, \beta_{3k}$$

即 $C_{\beta_{1k}}, C_{\beta_{2k}}, C_{\beta_{3k}}$ 中向量线性无关,

则 $\tilde{C}_{\beta_{1k}}, \tilde{C}_{\beta_{2k}}, \tilde{C}_{\beta_{3k}}$

(证明类似)

例 $J = \begin{pmatrix} 0 & 1 & & & \\ & 0 & & & \\ & & 0 & 1 & \\ & & & 0 & \\ & & & & 0 & 1 \\ & & & & & 0 \end{pmatrix}$

令 $e_i = \begin{pmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix} \leftarrow$ 第 i 位置是 "1".

考虑 $\left. \begin{aligned} v_1 &= e_2 + e_1 \\ v_2 &= e_2 + e_3 \\ v_3 &= e_2 + e_5 \end{aligned} \right\} \text{线性无关}$

则 $Av_1 = Av_2 = Av_3 = e_1$

从而 $Av_1, v_1, Av_2, v_2, Av_3, v_3$

线性相关.

2. (1)

$$e^{As} \cdot e^{At}$$

$$= \left(I + As + \frac{A^2 s^2}{2!} + \dots \right) \left(I + At + \frac{A^2 t^2}{2!} + \dots \right)$$

$$= \left(\sum_{j=0}^{\infty} \frac{A^j}{j!} s^j \right) \cdot \left(\sum_{k=0}^{\infty} \frac{A^k}{k!} t^k \right)$$

$$= \sum_{j=0}^{\infty} \sum_{k=0}^{\infty} \frac{A^{j+k}}{j! k!} s^j t^k$$

令 $n = j + k \Rightarrow k = n - j$, 上式

$$= \sum_{j=0}^{\infty} \sum_{n=j}^{\infty} \frac{A^n s^j t^{n-j}}{j! (n-j)!}$$

注

$$= \sum_{n=0}^{\infty} \sum_{j=0}^n \frac{A^n}{n!} \cdot \frac{n!}{j! (n-j)!} s^j t^{n-j}$$

$$= \sum_{n=0}^{\infty} \frac{A^n (s+t)^n}{n!} = e^{A(s+t)}.$$

$$\text{注: } \left\{ (n, j) \mid \begin{array}{l} n \geq j, j \geq 0 \\ n, j \in \mathbb{Z} \end{array} \right\}$$

$$= \left\{ (j, n) \mid \begin{array}{l} j \leq n, n \geq 0 \\ j, n \in \mathbb{Z} \end{array} \right\}$$

$$(2) \wedge s=1, t=-1$$

$$e^A \cdot e^{-A} = e^{A[1+(-1)]} = e^{0_{n \times n}} = I_n$$

(3) 若 $AB=BA$, 则由定义,

$$Ae^{Bt} = e^{Bt}A$$

$$\hat{g}(t) = e^{(A+B)t} e^{-Bt} e^{-At}$$

注意 $\frac{d e^{At}}{dt} = A e^{At}$.

则

$$\begin{aligned} \frac{d g(t)}{dt} &= (A+B) e^{(A+B)t} e^{-Bt} e^{-At} \\ &+ e^{(A+B)t} (-B) e^{-Bt} e^{-At} \\ &+ e^{(A+B)t} e^{-Bt} (-A) e^{-At} = 0 \end{aligned}$$

$$\Rightarrow g(t) = C (\text{常数矩阵})$$

$$g(0) = C = I.$$

因此 $g(t) = I \quad \forall t$.

$$\Rightarrow e^{(A+B)t} e^{-Bt} e^{-At} = I_n$$

两边右乘 $e^{At} \cdot e^{Bt}$, 得

$$e^{(A+B)t} = e^{At} \cdot e^{Bt} \quad \forall t.$$

(3) 的另外证明:

定义矩阵范数 (上学期内容)

$$M_n(\mathbb{C}) \xrightarrow{\|\cdot\|} \mathbb{R}$$

$$A = (a_{ij}) \longmapsto \|A\| = \sqrt{\sum_{i,j=1}^n |a_{ij}|^2}$$

性质: $\|A\| \geq 0, \|A\| = 0 \Leftrightarrow A = 0$

$$\|A+B\| \leq \|A\| + \|B\|, \|AB\| \leq \|A\| \cdot \|B\|$$

$$\text{现设 } f(z) = e^z, f_p(z) = 1 + z + \frac{1}{2!}z^2 + \dots + \frac{1}{p!}z^p$$

$$AB = BA \Rightarrow f_p(A)f_p(B)$$

$$= \left(\sum_{k=0}^p \frac{1}{k!} A^k \right) \left(\sum_{l=0}^p \frac{1}{l!} B^l \right) \quad \wedge \quad k+l=m$$

$$= \sum_{m=0}^p \left(\sum_{k=0}^m \frac{m!}{k!l!} A^k B^{m-k} \right) \cdot \frac{1}{m!} + \sum_{k+l > p} \frac{A^k B^l}{k!l!}$$

$$= \sum_{m=0}^p \frac{1}{m!} (A+B)^m + \sum_{k+l > p} \frac{A^k B^l}{k!l!}$$

$$= f_p(A+B) + \sum_{k+l > p} \frac{A^k B^l}{k!l!}$$

$$\| f_p(A)f_p(B) - f_p(A+B) \|$$

$$\leq \sum_{m > p} \left(\sum_{k+l=m} \frac{\|A\|^k}{k!} \cdot \frac{\|B\|^l}{l!} \right)$$

$$= \sum_{m > p} \frac{(\|A\| + \|B\|)^m}{m!}$$

因为 $\sum_{m=0}^{\infty} \frac{1}{m!} (\|A\| + \|B\|)^m = e^{\|A\| + \|B\|}$

因此, 当 $p \rightarrow +\infty$, 不等式右边趋于零.

即 $\lim_{p \rightarrow +\infty} \|f_p(A)f_p(B) - f_p(A+B)\|$
 $= 0 = \|f(A)f(B) - f(A+B)\|$
 $\left(\lim_{p \rightarrow +\infty} f_p(A) = f(A), \lim_{p \rightarrow +\infty} f_p(B) = f(B) \right)$

3. 设 $P^T A P = J = \begin{pmatrix} J_1 & & \\ & \ddots & \\ & & J_s \end{pmatrix}$

$J_i = \begin{pmatrix} \lambda_i & & \\ & \ddots & \\ & & 1 \\ & & & \ddots & \\ & & & & \lambda_i \end{pmatrix}_{r_i \times r_i}$

$$\text{当 } \lambda_i = 1, \text{ 且 } r_i = 1 \quad \lim_{k \rightarrow +\infty} J_i^k = 1$$

$$\text{当 } |\lambda_i| < 1, \quad \lim_{k \rightarrow +\infty} C_k^l \lambda_i^{k-l} = 0$$

$$\Rightarrow \lim_{k \rightarrow +\infty} J_i^k = O_{r_i \times r_i}$$

$$\text{因此 } \lim_{k \rightarrow +\infty} A^k \text{ 存在} \Leftrightarrow$$

$$|\lambda_i| < 1 \text{ 或 } \lambda_i = 1, \text{ 但 } r_i = 1, i=1, \dots, s$$

$$\text{极限矩阵} = P \begin{pmatrix} \overset{1}{\ddots} \overset{m}{\ddots} & & \\ & 0 & \\ & & \ddots & 0 \end{pmatrix} P^{-1}$$

$$m = A \text{ 的特征值 } 1 \text{ 的代数重数}$$

4. 由定义

$$\sin A = \frac{e^{iA} - e^{-iA}}{2i}$$

$$\cos A = \frac{e^{iA} + e^{-iA}}{2}$$

$$\sin 2A = \frac{1}{2i} (e^{i2A} - e^{-i2A})$$

因为 e^{iA} 与 e^{-iA} 交换 (*)

$$\Rightarrow e^{i2A} = (e^{iA})^2, e^{-i2A} = (e^{-iA})^2$$

$$\Rightarrow \sin 2A = \frac{1}{2i} [(e^{iA})^2 - (e^{-iA})^2]$$

$$\stackrel{(*)}{=} \frac{1}{2i} (e^{iA} + e^{-iA}) \cdot (e^{iA} - e^{-iA})$$

5. 设 $P^{-1}AP = J = \begin{pmatrix} J_1 & & \\ & \ddots & \\ & & J_k \end{pmatrix}$

$$J_i = \begin{pmatrix} \lambda_i & 1 & & \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ & & & \lambda_i \end{pmatrix}_{r_i \times r_i}$$

因为 A 可逆, $\lambda_i \neq 0 \quad i=1, \dots, k$.

取定 $\mu_i \neq 0, \mu_i^2 = \lambda_i$ (μ_i 不唯一).

$$\text{令 } J_i^* = \begin{pmatrix} \mu_i & 1 & & \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ & & & \mu_i \end{pmatrix}_{r_i \times r_i}$$

Claim: $(J_i^*)^2$ 相似于 J_i

因为 $J_i^* = \mu_i I_{r_i} + N$

例 设 $A = \begin{pmatrix} 0 & 4 & 23 \\ 0 & 0 & 2020 \\ 0 & 0 & 0 \end{pmatrix}$

则不存在 $B \in M_3(\mathbb{C})$, $B^2 = A$

因为若存在 B , $B^2 = A$,

则 $r(B) = 2 = r(A)$ $B \neq 0$.

B 的 Jordan 标准型可能:

$$\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \quad \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

$\Rightarrow r(B^2) = 0$ 或 1 矛盾!

$$\begin{aligned}
 (J_i^*)^2 &= \mu_i^2 I_{r_i} + 2\mu_i N + N^2 \\
 &= \lambda_i I_{r_i} + 2\mu_i N + N^2.
 \end{aligned}$$

$$\Rightarrow \text{rank}((J_i^*)^2 - \lambda_i I_{r_i}) = r_i - 1$$

$\Rightarrow (J_i^*)^2$ 只有一个特征值 λ_i , 相应 Jordan 块只有一个.

$$\Rightarrow (J_i^*)^2 \text{ 相似于 } J_i \quad \text{令 } J^* = \begin{pmatrix} J_1^* & & \\ & \ddots & \\ & & J_k^* \end{pmatrix}$$

$$\text{令 } Q_i^{-1} (J_i^*)^2 Q_i = J_i \quad \text{令 } Q = \begin{pmatrix} Q_1 & & \\ & \ddots & \\ & & Q_k \end{pmatrix}$$

$$A = P J P^{-1} = P Q^{-1} (J^*)^2 Q P^{-1}$$

$$= [(Q P^{-1})^{-1} J^* (Q P)]^2$$

注: 平方根不唯一 $\Rightarrow B$ 不唯一.

6. 因为 $\text{rank } A = 1$

则 $A = uv^H$ $u, v \in \mathbb{C}^n$

Case-1 $v^H u \neq 0$, 则 A 相似于

$$J_1 = \begin{pmatrix} v^H u & & \\ & 0 & \\ & & \ddots \\ & & & 0 \end{pmatrix}$$

Case-2 $v^H u = 0$, 则 A 相似于

$$J_2 = \begin{pmatrix} 0 & 1 & & \\ & 0 & & \\ & & \ddots & \\ & & & 0 \end{pmatrix}$$

设 $P_1^{-1}AP_1 = J_1$ 或 $P_2^{-1}AP_2 = J_2$

则 $f(A) = P_1 f(J_1) P_1^{-1}$ 或 $P_2 f(J_2) P_2^{-1}$

Case-1.

$$f(J_1) = \begin{pmatrix} f(v^H u) & & \\ & f(0) & \\ & & \ddots \\ & & & f(0) \end{pmatrix}$$

找多项式 $p(z)$ 使得

$$p(v^H u) = f(v^H u) \quad p(0) = f(0)$$

即 Lagrange 插值多项式

$$p(z) = \frac{z - v^H u}{0 - v^H u} f(0) + \frac{z - 0}{v^H u - 0} f(v^H u)$$

$p(z)$ 和 $f(z)$ 在 A 谱上定义相同

$$\Rightarrow p(A) = f(A)$$

$$= \frac{A - v^H u I_n}{-v^H u} f(0) + \frac{A}{v^H u} f(v^H u)$$

$$= -\frac{f(0)}{v^H u} A + f(0) I_n + \frac{f(v^H u)}{v^H u} A$$

$$= f(0) I_n + \left(\frac{f(v^H u) - f(0)}{v^H u - 0} \right) u v^H$$

Case-2.

$$f(J_2) = \begin{pmatrix} f(0) & 1 \\ & 0 \\ & & f(0) \\ & & & \ddots \\ & & & & f(0) \end{pmatrix}$$

找多项式 $P(z)$, 使得

$$P(0) = f(0), \quad P'(0) = f'(0).$$

$$\text{令 } P(z) = f(0) + f'(0)z$$

$$P(A) = f(0)I_n + f'(0)A$$

7. 设 A 存在, 则有两种可能:

(1) A 可对角化,

$$P^{-1}AP = \begin{pmatrix} \lambda_1 & \\ & \lambda_2 \end{pmatrix}$$

$$\Rightarrow \sin A = P \begin{pmatrix} \sin \lambda_1 & 0 \\ 0 & \sin \lambda_2 \end{pmatrix} P^{-1}$$

但 $\begin{pmatrix} 1 & 2020 \\ 0 & 1 \end{pmatrix}$ 不可对角化, 矛盾!

(2) A 相似于 $\begin{pmatrix} \lambda_0 & 1 \\ 0 & \lambda_0 \end{pmatrix}$

$$P^{-1}AP = \begin{pmatrix} \lambda_0 & 1 \\ 0 & \lambda_0 \end{pmatrix}$$

$$\text{则 } \sin A = P \begin{pmatrix} \sin \lambda_0 & \cos \lambda_0 \\ 0 & \sin \lambda_0 \end{pmatrix} P^{-1}$$

$$\text{若 } \sin A = \begin{pmatrix} 1 & 2020 \\ 0 & 1 \end{pmatrix}$$

$$\text{则 } \sin \lambda_0 = 1 \Rightarrow \cos \lambda_0 = 0. \text{ 矛盾!}$$

注: 本题也可以使用

$$\sin^2 A + \cos^2 A = I_2$$

$$\Rightarrow \cos^2 A = \begin{pmatrix} 0 & 2 \times 2020 \\ 0 & 0 \end{pmatrix} \text{ 幂零阵.}$$

$$\Rightarrow \cos A \text{ 是 } 2 \text{ 阶幂零阵}$$

$$\Rightarrow \cos^2 A = 0 \text{ 矛盾!}$$